MANYARA REGION

FORM FOUR MOCK EXMAINATION

032/2A **CHEMISTRY 2A**

ACTUAL PRACTICAL A

(*For both school and private candidates*)

TIME: 2:30 HOURS **YEAR 2021**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions
2. Each question carries **twenty five (25)** marks
3. Qualitative analysis guide sheet authorized by NECTA and non- programmable calculators may be used.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room
5. Write your examination number on every page of your answer sheets
6. The following constant may be used.

   Na = 23, H = 1, S = 32, O = 16, Cl = 35.5, C = 12, N = 14

wazaelimu.com

1. You are provided with the following solutions:

   Q: Contains 3.65 g per litre of hydrochloric acid.

   P. Contains 4g of XOH in a litre of the solution

   MO: Methyl orange indicator

   Procedure:

   Pipette 20cm^3 or 25cm^3 of **P** into a titration flask. Add 2 drops of methyl orange indicator

   Titrate solution **Q** against **P** until indicator changes its colour and note the burette reading.

   Repeat the procedure to obtain three titre values.

Questions:

(a) (i) ……cm^{3}of **P** required ……cm^3 of **Q** for complete reaction.

(ii) Write a balanced chemical equation for the reaction between **Q** and **P**.

(iii) A colour of an indicator changed from …….. to ……

(b) Showing your procedures clearly, calculate the:

(i) Molarity of solution **Q**

ii) Molarity of solution **P**

iii) The molar mass of **P**

(c) (i) Relative atomic mass of **X**.

(ii) What is the name of element X.

2. (a) You are provided with sample R containing one cation and one anion. Carry out the guided systematic procedure in the table below to identify the cation and anion present in the sample R.

Table

S/N	Experiment	Observation	inference
(a)	Observe sample **R**		
(b)	Take 0.5 g of sample **R** into a test tube and add 3mls of distilled water		
(c)	Transfer 0.5 g of solid sample **R** in a test tube and add 3 drops of concentrated H_2SO_4 acid.		
(d)	Transfer about 0.5 g of sample **R** in a dry test tube and heat gently		
(e)	Dissolve Sample R in distilled water then divide the resulting solution into three portions.		
	(i) Add sodium hydroxide solution drop-wise until in excess into the first portion.		
	(ii) Add barium chloride solution followed by dilute HCl into the second portion**.**		
	(iii) Add drops of potassium hexacyanoferrate (III) into the third portion		

Conclusion

(i) The cation in sample **R** is …….

(ii) The anion in sample **R** is ……..

(iii) The name of sample **R** is ……..

(iv) The chemical formula of sample **R** is …….

(b) What happened when sample solution R reacts with barium Chloride?